

21CSE312P SOFTWARE ENGINEERING IN ARTIFICIAL INTELLIGENCE

Project Report

Register No. : RA2311026050166

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Semester : VI Semester

Programme : B.Tech. CSE AIML C

Academic Year : **2025-2026**



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
TIRUCHIRAPPALLI
COLLEGE OF ENGINEERING AND TECHNOLOGY

REGISTER NO. RA2311026050166

BONAFIDE CERTIFICATE

Certified as the bonafide record of work done by
_____ Afeef Ahmed Shaik _____, Register No.
_____ RA2311026050166 _____ of _____ VI/III _____ (Semester/Year),
B.Tech. _____ CSE - AIML - C _____ Programme in the project-based
course **21CSE312P, Software Engineering In Artificial Intelligence** at **SRM**
Institute of Science and Technology, Tiruchirappalli Campus during the Academic
Year 2025-2026.

Faculty In-charge

Head of the Department

Submitted for the End Semester Examination held on 05-04-2026

Examiner-1

Examiner-2

SRM Institute of Science and Technology
Faculty of Engineering and Technology
School of Computing
Department of CSE(AIML)
VISION AND MISSION OF THE DEPARTMENT

Vision

To build a world-renowned academic platform in Computational Intelligence by providing unique learning and research experiences in collaboration with industries and world-class universities.

Mission

1. To envision in creating, acquiring, and disseminating engineering knowledge on computational intelligence to elevate a student into a professional by imparting knowledge on mathematics, computing sciences, artificial intelligence, and software engineering along with the skills of cognitive computing.
2. To offer a unique learning environment through world class faculty, curriculum, modernized lab facilities, and an interactive classroom environment with real-time experience from industrial experts that leads to a computing career in the latest technologies.
3. To uplift the innovative research and development in computational intelligence and its allied fields by collaborating with renowned academic institutions and industries.
4. To produce graduates who are global innovators and leaders in the development of computational intelligence-based systems, along with the commitment to ethical responsibilities and lifelong learning.

Program Educational Objectives (PEO)

PEO - 1 Graduates will be able to perform in technical/managerial roles ranging from design, development, problem solving to production support in software industries and R&D sectors

PEO - 2 Graduates will be able to successfully pursue higher education in reputed institutions.

PEO - 3 Graduates will have the ability to adapt, contribute and innovate new technologies and systems in the key domains of Computer Science and Engineering.

PEO - 4 Graduates will be ethically and socially responsible solution providers and entrepreneurs in Computer Science and other engineering disciplines.

PEO - 5 Graduates will possess the additional skills in core computer science discipline with knowledge of Hardware, Software, Programming, Logic & Reasoning.

Mission of the Department to Program Educational Objectives (PEO) Mapping

18. (c) Mission of the Department to Program Educational Objectives (PEO) Mapping

	Mission Stmt. - 1	Mission Stmt. - 2	Mission Stmt. - 3	Mission Stmt. - 4	Mission Stmt. - 5
PEO - 1	3	-	-	-	1
PEO - 2	-	-	2	-	-
PEO - 3	-	3	3	-	-
PEO - 4	-	-	-	2	3
PEO - 5	-	-	3	3	3

3 – High Correlation, 2 – Medium Correlation, 1 – Low Correlation

Program Outcomes as defined by NBA (PO)

Engineering Graduates will be able to:

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO – Program Specific Outcomes (PSO)

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PSO - 1	<i>To understand, analyze, design, and develop computing solutions by applying fundamental concepts of computer science and engineering.</i>
PSO - 2	<i>To apply computing principles, skills and practices to develop solutions using logical and reasoning skills, for real life problems.</i>
PSO - 3	<i>Ability to utilize Artificial Intelligence and Machine learning principles to design and develop cutting edge solutions for meeting the current demand of the industry.</i>

Mapping Program Educational Objectives (PEO) to Program Learning Outcomes (PLO)

18. (d) Mapping Program Educational Objectives (PEO) to Program Outcomes (PO)

	Program Outcomes (PO)												Program Specific Outcomes (PSO)		
	1	2	3	4	5	6	7	8	9	10	11	12	PSO-1	PSO-2	PSO-3
	Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning			
PEO - 1	3	-	-	1	-	2	-	3	2	3	-	-	3	-	-
PEO - 2	-	2	2	-	3	-	3	2	-	-	-	-	-	-	3
PEO - 3	-	3	3	2	-	-	-	-	-	2	-	3	-	3	-
PEO - 4	-	2	3	-	-	3	2	-	2	-	2	2	-	-	3
PEO - 5	-	3	3	-	3	-	-	-	3	-	3	3	-	2	3

3 – High Correlation, 2 – Medium Correlation, 1 – Low Correlation

COURSE DESIGN

Course Learning Outcomes (CLO):	At the end of this course, learners will be able to:
CO – 1	Distinguish the different components in GPU systems
CO – 2	Create environments to work with different NGC container packages
CO – 3	Implement codes using jupyter notebook and pytorch
CO – 4	Develop and work with CUDA
CO – 5	Visualize different DL deployments for various scientific applications

CLO-PO Mapping

CLO	Program Outcomes (PO)														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	Engineering Knowledge	Problem Analysis	Design & Development	Conduct investigations of complex problems	Modern Tool Usage	Society & Culture	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO - 1	PSO - 2	PSO - 3
CO1	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	2	3	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	3	3	-	-	-	-	-	-	-	-	-	-	2	3
CO4	-	3	3	-	3	-	-	-	-	-	-	-	-	-	3
CO5	-	2	-	-	3	-	-	-	-	-	-	-	-	-	-

Report Evaluation:

Set 1:

Test Components	Date	Artifacts	Total Marks
1	12-02-2026	a. Installation of Oracle Virutal Box & Docker b. Load Ubuntu as Virtual Machine	10
2	18-02-2026	Installation and Pulling Specific NGC Packages - Both Usecases and Frameworks	10
3	20-02-2026	Pull and Push - Docker Hub	10
4	27-02-2026	Implementing Kubernetes in Mini-Kube	10
5	27-03-2026	Working with SLURM - Scripting & Kubeflow	10

Set 2:

Test Components	Date	Artifacts	Total Marks
6	08-04-2026	a. Using Jupyter note and PyTorch b. Run a simple data centric application using Jupyter note.	10
7	21-04-2026	Testing the GPGPU Computing	10
8	25-04-2026	a. Testing and working with Tensor (CUDA-X) b. Implementing Mixed Precision and Quantization Aware training	10
9	28-04-2026	Accelerating Neural Network Inferencing: TensorRT & Triton Inference Server	10
10	01-05-2026	a. Monitoring load Balancers & Schedulers b. Deployment of various services for monitoring, jupyter environment and other services	10

S. NO.	Component	Mark Calculation
1	Set 1	50/5 = 10
2	Set 2	50/5 = 10
	Total Marks	20

Mark Split Rubrics

S.no.	Criteria	Marks
Part 1: Continuous Assessment (Theory)		
1	CLA – 01 (10) & Quiz 1 (5)	15
2	CLA – 02 (10) & Quiz 2 (5)	15
Part 2: Experiments (Practice)		
3	Set 1 (Exp 1-5)	10
4	Set 2 (Exp 6-10)	10
Part 3: Software Paper Preparation		
5	Product Innovation	5
6	Paper as per Software Impacts Template	5
7	Turnitin Report (Copy Plagiarism & AI Plagiarism Report)	5
8	Demo Video as per Guidelines	5
Part 4: Presentation and Review		
9	GitHub Repo and Fork	5
10	Review 1 (Quality of the modules developed)	5
11	Review 2 (Containerization and Deployment)	10
12	Paper Acceptance from Journal or Conference	10
Extra-Mile Certification		
13	Coursera / NPTEL	10

Part 2 : Evaluation Sheet

Expt. No.	Date of Experiment	Name of the Experiment	Page No.	Marks (10)	Signature of Faculty
1	12-02-2026	Environment Setup	1		
2	18-02-2026	Working Docker Image and NGC Container	5		
3	20-02-2026	Installation and Pulling Packages from Docker Hub	20		
4	27-02-2026	Implementing and executing Kubernetes	30		
5	27-03-2026	Working with SLURM – Scripting & Kubeflow	56		
6	08-04-2026	Using Jupyter Notebook and Pytorch for Tensor Operation	65		
7	21-04-2026	Testing the GPGPU Computing	70		
8	25-04-2026	Testing CUDA-X Tensor Operation	74		
9	28-04-2026	Accelerating Neural Network Inferencing : TensorRT & Triton Inference Serve	81		
10	01-05-2026	Monitoring Load Balancers & Schedulers	88		
Total Marks					

Part No.	Paper Part (Refer Rubric Table)	Page No.	Marks	Signature of Faculty
1	Part 3 - 20 Marks			
2	Part 4 - 30 Marks			
Total Marks				